



Women's Boots Shoe Assembly

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Background

1.1 Context

This dataset models the **assembly of women's boots** (PU/Microfiber Upper + TPR/EVA/Rubber Outsole) based on **primary data** collected from a Chinese shoe manufacturer (Wenzhou) by the Carbonfact Collective initiative.

Unlike ecoinvent-sourced assembly proxies, this inventory captures the actual process steps, chemical inputs (from MSDS/SDS documents), electricity consumption (measured per machine), and material loss rates as reported by the factory.

1.2 Scope

- **Functional unit:** 1 pair of assembled women's boots
- **System boundary:** Assembly only (Cutting → Stitching → Lasting). Upstream material production (upper, outsole) is excluded and must be linked separately.
- **Background database:** ecoinvent 3.12 (cut-off)
- **Location variants:** GLO (global average), CN (China), IT (Italy)
- **Input/Output Ratios:** Cutting 1.0382, Stitching 1.006, Lasting 1.004 → **Aggregated IOR: 1.049**

1.3 Data Collection and Source Notes

- **Primary data collection:** Foreground data were collected directly from the supplier through a structured LCA data template and iterative follow-up exchanges.
- **Measured process data:** The dataset includes machine/process-level electricity consumption, process chemicals, and wastage information for cutting, stitching, and lasting.
- **Supporting documentation:** Adhesives, primer, cleaner, and oil paint were refined using supplier records plus MSDS/SDS or technical datasheets where available.
- **Modeling approach:** The inventories were modeled in Brightway2.5 and linked to ecoinvent 3.12 background datasets for electricity supply and waste treatment.
- **Gap filling:** Where exact background matches were unavailable, conservative proxies were used for selected chemical constituents and generic waste routing assumptions.

Analysis

2.1 Key Findings from Literature

Shoe assembly is generally a low-impact stage relative to material production (leather, synthetic upper, rubber outsole). However, the electricity mix and chemical inputs (adhesives, primers, solvents) can be significant in certain geographies.

2.2 Data Quality and Uncertainty Tiers

- **Tier 1 (primary):** Electricity consumption (measured), chemical quantities (from factory records), loss rates (measured)
- **Tier 2 (MSDS-derived):** Chemical compositions of glues, primers, cleaners, oil paint — modeled from Safety Data Sheets with ecoinvent proxies
- **Tier 3 (ecoinvent proxy):** Chemical factory infrastructure, waste treatment

2.3 Cross-Validation Summary

The supplier-specific dataset (ecoinvent 3.12, EF 3.1) is compared below against generic assembly datasets available in EF 3.1 for open-toed shoes, close-toed shoes, and boots. Source: [Shoe Assembly \(PU/Microfiber Upper + TPR/EVA/Rubber Outsole\) – Women's Boots](#)

Indicator	Unit	Supplier GLO (ei 3.12)	Supplier CN (ei 3.12)	Supplier IT (ei 3.12)	Generic Open-Toed (EF 3.1)	Generic Close-Toed (EF 3.1)	Generic Boot (EF 3.1)
GHG	kg CO ₂ eq	1.15	1.38	0.81	2.17	2.32	2.36
ACD	mol H ⁺ eq	5.62e-3	7.68e-3	3.15e-3	0.0132	0.0142	0.0145
ETF	CTUe	7.11	8.34	5.63	52.86	56.48	56.81
FRU	MJ	16.86	16.88	13.85	29.08	31.66	32.38
FWE	kg P eq	4.85e-4	2.68e-4	2.18e-4	2.68e-5	3.61e-5	3.82e-5
HTC	CTUh	4.04e-10	3.98e-10	3.73e-10	6.12e-9	6.80e-9	6.82e-9
HTN	CTUh	7.81e-9	9.04e-9	4.79e-9	3.98e-8	4.14e-8	4.17e-8
IOR	kBq U ²³⁵ eq	0.127	0.061	0.077	0.115	0.125	0.128
LDU	Pt	2.43	3.08	2.04	7.65	8.07	8.15
MRU	kg Sb eq	2.92e-6	2.89e-6	3.06e-6	4.37e-6	5.32e-6	5.52e-6
OZD	kg CFC-11 eq	3.50e-8	3.19e-8	4.23e-8	1.85e-8	2.35e-8	2.36e-8
PCO	kg NMVOC eq	3.97e-3	5.20e-3	2.71e-3	0.00912	0.00978	0.00993
PMA	disease inc.	6.73e-8	1.18e-7	3.23e-8	4.25e-7	4.44e-7	4.49e-7
SWE	kg N eq	1.70e-3	2.17e-3	1.15e-3	0.00239	0.00262	0.00267
TRE	mol N eq	1.24e-2	1.80e-2	7.02e-3	0.0242	0.0264	0.0269
WTU	m ³ eq	0.283	0.205	0.305	0.889	1.031	1.067

The supplier-specific dataset shows consistently lower impacts across most categories compared to generic EF 3.1 assembly datasets. For GHG, the GLO variant (1.15 kg CO₂ eq/pair) is **51% lower** than the generic boots dataset (2.36 kg CO₂ eq/pair). Freshwater ecotoxicity and human toxicity are an order of magnitude lower, reflecting the value of primary data over generic proxies.

Notable exceptions: **freshwater eutrophication (FWE)** is ~10x higher in the supplier dataset, likely due to the electricity mix contribution; **ozone depletion (OZD)** is also slightly higher. These warrant further investigation but may reflect different system boundary assumptions in the generic EF 3.1 datasets.

Process Inventory

3.1 Summary Table — All 8 Datasets

Process	Locations	FU	Key Inputs
Cutting	GLO, CN, IT	1 pair	Electricity (0.02 kWh), Oil paint (0.013 kg), Waste PU
Stitching	GLO, CN, IT	1 pair	Electricity (0.1 kWh), 163N glue (0.0006 kg), Waste PU
Lasting	GLO, CN, IT	1 pair	Electricity (1.2 kWh), PU adhesive (0.02 kg), Primer (0.02 kg), Cleaner (0.006 kg)
Shoe Assembly (top-level)	GLO, CN, IT	1 pair	Cutting + Stitching + Lasting with cumulative loss rates

Process	Locations	FU	Key Inputs
163N stitching glue	GLO, CN, IT	1 kg	Organic solvent (0.9 kg), Natural rubber seal (0.1 kg)
BW553 primer	GLO, CN, IT	1 kg	Ethyl acetate (0.6 kg), Acetone (0.25 kg), Solvent (0.1 kg), PU foam (0.05 kg)
Shoe surface cleaner	GLO	1 kg	Butanone proxy (0.52 kg), Acetone (0.35 kg), Xylene (0.1 kg)
Oil paint for print	GLO	1 kg	PP resin (0.4 kg), Cyclohexanone (0.4 kg), Organic pigment proxy (0.18 kg)

Methodological Choices

4.1 Chemical/Material Proxies

- **Butanone** → [market for solvent, organic](#) (no specific ecoinvent dataset)
- **Organic pigment** → [market for solvent, organic](#) (conservative proxy)
- **Silicone oil** → [market for silicone product](#)
- **Natural rubber component** → [market for seal, natural rubber based](#)
- **PU foam in primer** → [polyurethane production, flexible foam, MDI-based](#)

4.2 Electricity

- GLO/CN: [market group for electricity, medium voltage](#)
- IT: [market for electricity, medium voltage](#)
- Chemical sub-processes (IT/CN variants): same pattern; GLO uses market group

4.3 Loss Rates

Per-sub-process Input/Output Ratios (Cutting 1.0382, Stitching 1.006, Lasting 1.004) cascade in the aggregate **Shoe Assembly** activity as explicit amount multipliers (Lasting $\times 1$, Stitching $\times 1.004$, Cutting $\times 1.010024$), so the caller sees the real processing effort per pair of final assembly.

Per the harmonised waste-handling convention (see Section 4.5), **no cumulative** [market for waste yarn and waste textile](#) **flow is present at the aggregate level**. Material waste is the responsibility of the upstream material-supplier dataset: the caller applies the cumulative IOR = $1.0382 \times 1.006 \times 1.004 = 1.049$ at the material-input layer, so that the treatment of cut-offs/trim is counted exactly once — where the material is supplied.

4.4 ecoinvent 3.12 Update Notes

- [market for polypropylene, granulate](#) changed from GLO → RoW (no GLO market in 3.12)
- All other activity names unchanged from 3.10
- Format version updated from [cutoff 11](#) to [cutoff 12](#)

4.5 Waste Handling

Process-related waste is split into two categories with distinct modelling rules, harmonised across Carbonfact's textile/footwear LCI datasets:

- **Auxiliary / chemical waste** — cured polyurethane adhesive / primer residues, off-spec chemicals, spent cleaners, contaminated solvents. Kept **inline inside each sub-process** as a negative-amount exchange to the appropriate ecoinvent waste-treatment activity ([market for waste polyurethane](#), -0.01 kg per pair in Cutting / Stitching / Lasting). These flows are **process-specific** and cannot be inferred from a material loss rate.
- **Textile / footwear material waste** — cut-offs of upper fabric, microfiber, PU sheet, EVA/TPR/rubber scraps, outsole trim. **Excluded from the process datasets**. The [market for waste yarn and waste textile](#) exchange is kept at **0** in each sub-process (for documentation only) and **no cumulative flow is aggregated** at the top-level Shoe Assembly activity. Material waste is handled **downstream via the IOR chain** (Section 4.3), so the caller's material-supplier dataset carries the treatment burden for scraps exactly once.

This separation keeps process chemistry and energy intensities comparable across sites and electricity mixes, and makes material-loss accounting auditable and free of double counting.

LCIA Results (EF 3.1, ecoinvent 3.12)

Functional unit: **1 pair of assembled women's boots**. All values per pair.

Indicator	Unit	GLO	CN	IT
GHG	kg CO2 eq	1.154	1.377	0.807
ACD	mol H+ eq	5.62e-3	7.68e-3	3.15e-3
ETF	CTUe	7.11	8.34	5.63
FRU	MJ	16.86	16.88	13.85
FWE	kg P eq	4.85e-4	2.68e-4	2.18e-4
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TRE	mol N eq	1.24e-2	1.80e-2	7.02e-3
WTU	m3 world eq	0.283	0.205	0.305

DQR Scores (PEF, GHG-weighted)

Location	P	TiR	TeR	GR	DQR
GLO	1.31	2.15	2.15	2.15	1.94
CN	1.29	2.14	2.14	2.14	1.93
IT	1.36	2.18	2.18	2.18	1.98

GWP Contribution — GLO

Lasting dominates (~88%) due to 1.2 kWh electricity. Stitching contributes ~6.6%, Cutting ~5.3%. Cumulative material waste is no longer a line item in the aggregate — it is handled downstream via the IOR chain (see Section 4.3 / 4.5).

Limitations and Open Items

Item	Status	Next Step
Chemical proxies (butanone, pigment)	Conservative proxy used	Refine if specific datasets become available
Single factory data	Wenzhou factory only	Validate against additional manufacturers
Waste treatment routing	Generic waste PU/textile	Confirm actual waste streams at factory

References

- ecoinvent 3.12 cut-off database
- Factory data collection sheets (Wenzhou, China)
- MSDS: BW553 Lasting Primer, 163N Stitching Glue, Shoe Surface Cleaner, Oil Paint for Print